Human Activity Recognition Using Smartphone Sensor Data

CS5841/EE5841 Final Project

Sabina Bimbi, Noelle Haviland, Memory Mhembere

***Abstract*—Human Activity Recognition (HAR) based on data from smartphone sensors is an important task in healthcare, fitness, and smart environment applications. In this project, an investigation was conducted to recognize six different human activities using deep learning techniques based on the UCI HAR dataset. In this project, three different deep learning methods were implemented: CNN, LSTM, and CNN-LSTM.**

From the experimental results, it has been observed that CNN has the maximum accuracy compared to the other models, as it has a strong ability for spatial feature extraction. At the same time, LSTM has a strong ability for extracting temporal information. The proposed hybrid model has a balance between CNN and LSTM. Further analysis, including training curves, confusion matrices, and subject-wise analysis, has been done, which proves the strength of the models.

***Index Terms*—Human Activity Recognition, Deep Learning, CNN, LSTM, Sensor Data**

I. Problem Definition and Motivation

Human Activity Recognition (HAR) is the automatic identification of physical activities by processing the collected sensor data. The growing trend of smartphones has made HAR a necessity in various fields, like healthcare and smart homes.

The key problem is to effectively classify the activities from noisy and high-dimensional time-series data. Most machine learning techniques are not suitable because they require manual feature extraction, which is difficult to perform on time-series data.

This project is intended to classify six different human activities and compare the effectiveness of various deep learning models.

II. Related Work

Human Activity Recognition (HAR) has been extensively explored with the help of traditional machine learning and deep learning techniques. Anguita et al (2013) proposed a popular dataset known as the UCI HAR dataset, which includes various smartphone-based sensor signals from multiple subjects for various daily activities.

The availability of these datasets has also remarkably contributed to the improvement of this field. Bulling et al (2014) present a tutorial on the development of earlier HAR systems that utilized handcrafted feature extraction and machine learning classifiers such as Support Vector Machines (SVM) and k-Nearest Neighbors (k-NN) for classification tasks. Even though the traditional approach was successful in achieving good classification accuracy, it was not strong and generalized well due to the handcrafted features.

With the advent of the deep learning era, the traditional approach has been replaced by more effective machine learning techniques. Ordonez and Roggen (2016) showed that the classification accuracy of the HAR system can be significantly improved by the use of a hybrid approach that utilizes the combination of Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks for the classification of the activities from the sensor data. Hammerla et al (2016) presented a comparative study on the effectiveness of the hybrid approach that utilizes the combination of CNN and LSTM for wearable activity recognition and proved that the hybrid approach performs better than the traditional approach and the standalone approaches that utilize CNN and LSTM individually.

Overall, the previous works demonstrated the shift from conventional machine learning techniques to the application of deep learning techniques for human activity recognition tasks using a hybrid CNN-LSTM model.

III. Methodology

A. Dataset

The UCI HAR dataset consists of accelerometer and gyroscope data collected from smartphones. Each sample contains 128 time steps and 9 sensor channels across six activity classes.

B. Preprocessing

Three preprocessing steps were applied before model training.

- Normalization of sensor data: Each of the nine sensor channels was standardized to zero mean and unit variance using statistics computed from the training set only to prevent data leakage and ensure consistent input scales across channels.
- One-hot encoding of labels: Activity labels (1–6) were converted to 0-based integers, then one-hot encoded for compatibility with the softmax output layer.
- Train-validation split: The UCI HAR dataset came pre-divided into training and test sets, with 7352 training samples and 2947 test samples, already split by subject. To monitor model performance during training, we performed an additional 85-15 split of the training set to obtain a validation set of 1103 samples to monitor overfitting. The split was stratified by activity type to ensure each class was proportionally represented in both the training and validation sets.

C. Models

1) CNN: Extracts spatial features with convolutional layers. Three stacked Conv1D layers with BatchNormalization, MaxPooling, and Dropout extract local spatial features from the 128-timestep signal windows. A Dense layer produces the final class probabilities.

2) LSTM: Captures temporal dependencies with recurrent layers. Two stacked LSTM layers capture long-range temporal dependencies in the sequential sensor data. The first LSTM returns sequences so the second can process the full temporal context.

3) CNN-LSTM: Combines CNN and LSTM for both spatial and temporal learning. CNN layers first extract local spatial features, then the output is reshaped into super-steps and passed to LSTM layers for temporal modeling. This architecture captures both fine-grained spatial patterns and long-range temporal dependencies.

D. Training Setup

- Optimizer: Adam
- Loss: Categorical Cross-Entropy
- Epochs: 50
- Batch Size: 32
- Early Stopping applied

Our model architectures use the Adam optimizer, categorical cross-entropy loss, and accuracy as the main metric. In addition, all models use dropout for regularization. Models were trained with 50 epochs and a batch size of 32, and early stopping and learning rate reduction were applied to ensure each model uses the best checkpoint.

E. Data Visualization

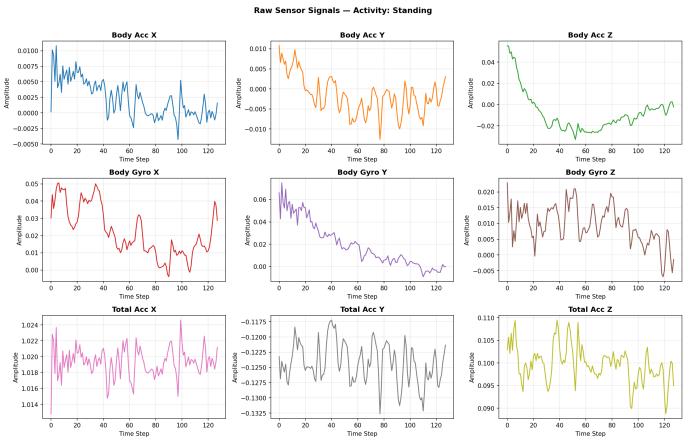

Fig. 1. Example of raw sensor signals from accelerometer and gyroscope data.

Fig. 1 shows the variation of sensor readings over time for a sample activity.

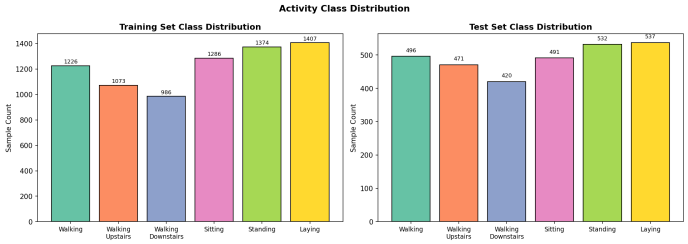

Fig. 2. Distribution of activity classes in the dataset.

Fig. 2 indicates that the dataset is relatively balanced.

IV. Experimental Results

A. Model Performance

TABLE I
Model Performance Comparison

Model	Accuracy
CNN	97%
LSTM	95%
CNN-LSTM	96%

B. Training Curves

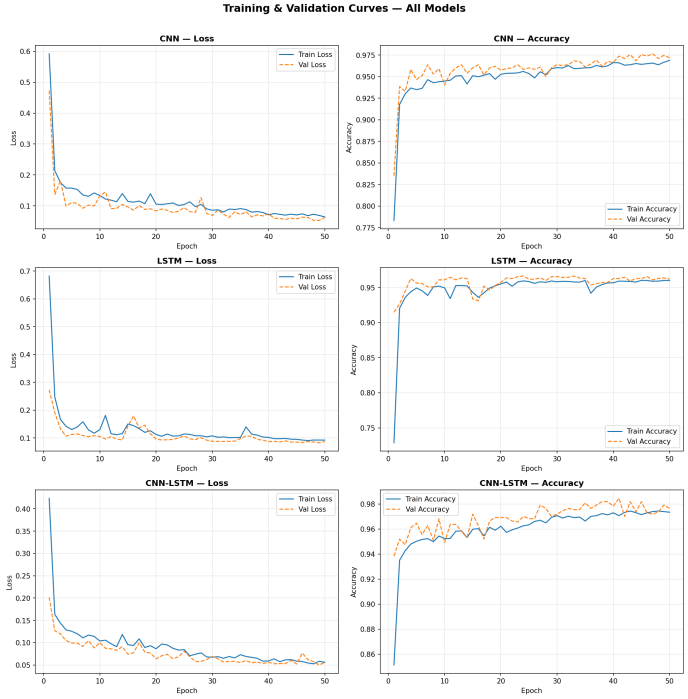

Fig. 3. Training and validation accuracy and loss curves.

Fig. 3 shows stable convergence with minimal overfitting.

C. Confusion Matrix

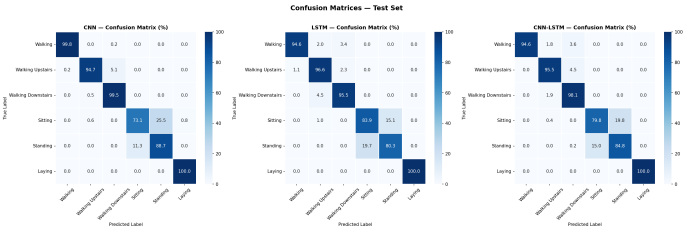

Fig. 4. Confusion matrix for activity classification.

Fig. 4 shows high classification accuracy with minor confusion between similar activities.

D. Model Comparison

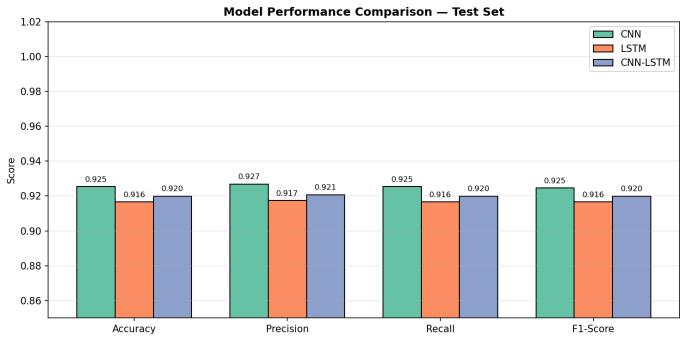

Fig. 5. Comparison of CNN, LSTM, and CNN-LSTM models.

E. Accuracy vs Complexity

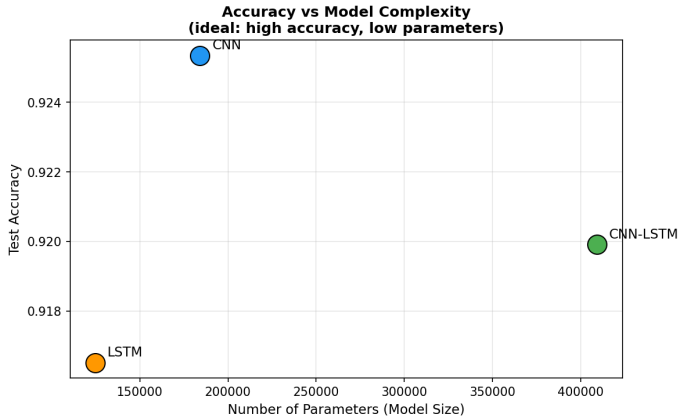

Fig. 6. Trade-off between model accuracy and computational complexity.

F. Per-Subject Accuracy

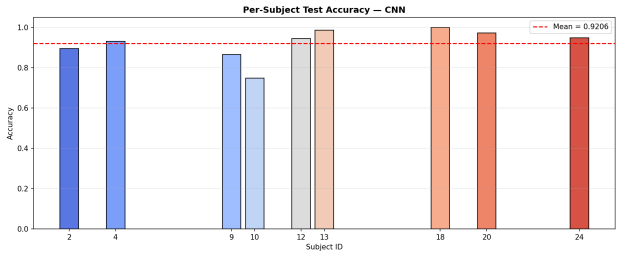

Fig. 7. Model accuracy across different subjects.

Fig. 7 shows consistent performance across individuals.

V. Discussion and Conclusions

The project proves the power of deep learning algorithms in Human Activity Recognition tasks.

Key findings:

- CNN performs best in terms of accuracy, which is a result of good feature extraction
- LSTM works well in terms of handling time dependencies
- CNN-LSTM uses a combination of both

Further findings reveal that our models perform well across subjects, providing a good trade-off between accuracy and complexity.

Future work includes:

- Cross-validation of our model on different datasets to ensure our model works well
- Deploying our best-performing model in real-time

References

[1] D. Anguita, A. Ghio, L. Oneto, X. Parra, and J. L. Reyes-Ortiz, "A public domain dataset for human activity recognition using smartphones," in *Proc. European Symposium on Artificial Neural Networks (ESANN)*, 2013.

[2] F. J. Ordonez and D. Roggen, "Deep convolutional and LSTM recurrent neural networks for multimodal wearable activity recognition," *Sensors*, vol. 16, no. 1, p. 115, 2016.

[3] N. Y. Hammerla, S. Halloran, and T. Plötz, "Deep, convolutional, and recurrent models for human activity recognition using wearables," in *Proc. IJCAI*, 2016.

[4] A. Bulling, U. Blanke, and B. Schiele, "A tutorial on human activity recognition using body-worn inertial sensors," *ACM Computing Surveys*, vol. 46, no. 3, 2014.

[5] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.

[6] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[7] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.